

Quantum Communications and Networks(QU.50013)

Exercise Set 2

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Problem 1

(a)

$$\rho_{AB} = \begin{pmatrix} \cos^2 \theta & 0 & 0 & \sin \theta \cos \theta \\ 0 & 0 & 0 & 0 \\ \sin \theta \cos \theta & 0 & 0 & \sin^2 \theta \end{pmatrix}$$

$$\Rightarrow \rho_A = \text{Tr}_B \rho_{AB} = \begin{pmatrix} \text{Tr} \begin{pmatrix} \cos^2 \theta & 0 \\ 0 & 0 \end{pmatrix} & \text{Tr} \begin{pmatrix} 0 & \sin \theta \cos \theta \\ 0 & 0 \end{pmatrix} \\ \text{Tr} \begin{pmatrix} 0 & 0 \\ \sin \theta \cos \theta & 0 \end{pmatrix} & \text{Tr} \begin{pmatrix} 0 & 0 \\ 0 & \sin^2 \theta \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \cos^2 \theta & 0 \\ 0 & \sin^2 \theta \end{pmatrix}$$

(b) ρ_B is in its diagonal form in the computational basis, so its eigenvalues are $\cos^2 \theta$ and $\sin^2 \theta$.

$$\Rightarrow S(\rho_A) = -\cos^2 \theta \log_2 \cos^2 \theta - \sin^2 \theta \log_2 \sin^2 \theta$$

(c) $S(\rho_A)(\theta = 0) = 0$, $S(\rho_A)(\theta = \pi/4) = 1$. In fact, the former corresponds to the classical state $|00\rangle$, while the latter corresponds to the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

(d) *If the global state is known to be pure*, then the fact that ρ_A describes a mixed state could allude to global entanglement, although no local operation nor measurement can distinguish a mixed state from a subsystem of an entangled state with identical reduced density matrices.

Problem 2

(a) $E(0) = -1$, $E(\pi/3) = -1/2$, $E(\pi/2) = 0$, $E(\pi) = +1$

(b) $\theta = 0$ corresponds to perfect anticorrelation, i.e., measurement outcomes are always opposite. As θ increases to $\pi/2$, E increases to zero, meaning a net anticorrelation but with less and less significance. At $\theta = \pi/2$, $E = 0$ implies the absence of correlation. After that point, the measurement outcomes are positively correlated, with more and more significance as θ increases to π . Finally, at $\theta = \pi$, the measurements are perfectly correlated, i.e., always identical outcomes.

(c) If Bob measures along z , Alice can be absolutely certain that Bob will obtain -1 . Along x , Alice cannot make any prediction about Bob's outcome.

(d) These correlations do not allow superluminal information transfer, as these correlations can only be compared and utilized once the two measurement outcomes have been compared at a common point in space.

Problem 3

(a)

$$C = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

(b)

$$CC^\dagger = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{2}}{3} \\ \frac{\sqrt{2}}{3} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{3+2\sqrt{2}}{6} & \\ & \frac{3-2\sqrt{2}}{6} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

Thus, the eigenvalues of $CC^\dagger$ are $(3 \pm 2\sqrt{2})/6$, meaning the Schmidt coefficients are $(\sqrt{2} \pm 1)/\sqrt{6}$.

(c) Because the bipartite system is in a pure state, the Schmidt rank is an exact criterion for separability. Since there are two nonzero Schmidt coefficients, therefore, the system is entangled. In fact, the state can be written as:

$$|\psi\rangle = \frac{\sqrt{2}+1}{\sqrt{6}} |+\rangle |+\rangle + \frac{\sqrt{2}-1}{\sqrt{6}} |-\rangle |-\rangle.$$

(d) Local unitary operations act independently on each product state in the Schmidt decomposition, yielding yet another product state. Thus, the Schmidt coefficients, and consequently the Schmidt rank, are left unchanged. This means that entanglement, as a physical resource, cannot be created nor destroyed by local operations.

Problem 4

(a) $E(a, b) = E(a, b') = E(a', b) = -\cos \frac{\pi}{4} = -\frac{1}{\sqrt{2}}$, $E(a', b') = -\cos \frac{3\pi}{4} = +\frac{1}{\sqrt{2}} \Rightarrow S = -2\sqrt{2}$

(b) The CHSH bound has been violated, implying that no local hidden-variable theory can account for the correlation described by quantum entanglement.

(c) $|S| = 2\sqrt{2}$, exactly tight with the Tsirelson bound.

(d) The visibility factor affects S by $S_{exp} = VS_{ideal}$, so for $S_{ideal} = -2\sqrt{2}$ to yield $|S_{exp}| > 2$, we need at least $V > 1/\sqrt{2}$. This threshold represents the minimum required fidelity of our engineering capabilities to meaningfully perform the abstract Bell test and determine whether local hidden variable theory can describe our microscopic universe.

Problem 5

(a) $CX |0\rangle |0\rangle = |0\rangle |0\rangle$, $CX |1\rangle |0\rangle = |1\rangle |1\rangle$

(b) $CX |+\rangle |0\rangle = (CX |0\rangle |0\rangle + CX |1\rangle |1\rangle)/\sqrt{2} = (|0\rangle |0\rangle + |1\rangle |1\rangle)/\sqrt{2} = |\Phi^+\rangle$

(c) $|+\rangle |+\rangle = (|0\rangle |0\rangle + |0\rangle |1\rangle + |1\rangle |0\rangle + |1\rangle |1\rangle)/2$ The two states are different.

(d) Orthogonal states can be conditionally copied; for the computational basis, CX is such a copier. The no-cloning theorem prohibits the cloning of arbitrary states, which CX cannot do, as demonstrated with $|+\rangle$.

Problem 6

(a) $\langle\psi|\phi\rangle = \langle\psi|\phi\rangle \langle 0|0\rangle = (\langle\psi|\langle 0|)(|\phi\rangle |0\rangle) = (\langle\psi|\langle 0|)U^\dagger U(|\phi\rangle |0\rangle) = (\langle\psi|\langle\psi|)(|\phi\rangle |\phi\rangle) = (\langle\psi|\phi\rangle)^2$

(b) $\langle\psi|\phi\rangle = 0, 1$. The former implies orthogonal states; latter identical states.

(c) $\langle 0|+\rangle = 1/\sqrt{2}$. Thus, no unitary U can clone both $|0\rangle$ and $|+\rangle$.

(d) The no-cloning theorem states that no unitary process can clone arbitrary non-orthogonal states perfectly. This prevents actions like signal amplification for arbitrary quantum channels.

Problem 7

(a) $\left(\langle\psi_1|^{\otimes N}\right)\left(|\psi_2\rangle^{\otimes N}\right) = (\langle\psi_1|\psi_2\rangle)^N = s^N$

(b) The overlap vanishes exponentially to zero as $N \rightarrow \infty$, meaning the two states become asymptotically orthogonal and thus distinguishable.

(c) While a single-sample state discrimination is impossible, it would be possible if one had an arbitrary cloner: clone sufficiently many copies of the given state, and perform the orthogonal state discrimination. Thus, the impossibility of non-orthogonal state discrimination implies that of arbitrary state cloning.

(d) $P_e(s = 1/2) = (2 - \sqrt{3})/4 \approx 0.067$. With a perfect cloner, one could clone the given state to get the refined error bound $P_e(s = 1/2^N) = (1 - \sqrt{1 - 4^{-N}})/2$, which approaches zero in the limit $N \rightarrow \infty$. Thus, a perfect cloning machine could violate the Helstrom error probability bound.